LLM Week 4

Consider a vocabulary $\mathcal{V}$,

$\mathcal{V}$ =([start], building, character, a, is, astronomy, science, experience, natural,[end]).

Assume that we have a pre-trained GPT model for text generation and the first input token to the model is always the special token [start]. The prediction probabilities outputted by the model are given below.

$$\begin{array}{c} \begin{array}{cccccccccc} s & b & c & a & is & a & s & e & n & e \end{array} \\ \hat{Y} = \begin{bmatrix} 0.05 & 0.02 & 0.14 & 0.07 & 0.09 & \boxed{0.41} & 0.08 & 0.05 & 0.08 & 0.01 \\ 0.10 & 0.04 & 0.06 & 0.16 & 0.43 & 0.01 & 0.06 & 0.09 & 0.01 & 0.04 \\ 0.05 & 0.07 & 0.28 & 0.29 & 0.03 & 0.08 & 0.04 & 0.08 & 0.04 & 0.04 \\ 0.10 & 0.02 & 0.48 & 0.02 & 0.01 & 0.07 & 0.06 & 0.14 & 0.07 & 0.01 \\ 0.08 & 0.29 & 0.04 & 0.01 & 0.04 & 0.03 & 0.25 & 0.05 & 0.15 & 0.06 \\ 0.14 & 0.22 & 0.13 & 0.08 & 0.01 & 0.03 & 0.06 & 0.23 & 0.08 & 0.03 \end{bmatrix} \end{array}$$

The zeroth row of the matrix is the output probability distribution by the model given the special [Start] token as input. Following the Greedy Search decoding strategy, the subsequent rows give the conditional probability distribution conditioned over the previous tokens. If you think the given information is insufficient for any sub-questions, then enter -1 as your answer

1) What is the probability of the predicted word equal to "Astronomy" given the "[Start]" token as the initial input $P(y_1 = Astronomy|y_0 = [Start])$?

0.41 ✓ 0.41

2) What is the probability of the sequence "Astronomy is a character building experience"?

Note: Enter your answer correct upto 4 decimal places.

-1

$0.41 \times 0.43 \times 0.29 \times 0.48 \times 0.29 \times 0.23 = 0.0016$

No, the answer is incorrect.
Score: 0

Accepted Answers:

✓ (Type: Range) 0.001,0.002

2 points

3) What is the probability of the sequence " Astronomy is a natural science "?

-1

$0.41 \times 0.43 \times 0.29 \times 0.07 \times 0.25 = 0.0009$

Yes, the answer is correct.
Score: 1

Accepted Answers:

(Type: Numeric) 0 ✓
(Type: Numeric) -1

1 point

4) Suppose we use Top$-k$ sampling, with $k = 3$, What is the probability of the predicted word equal to "Astronomy" given the "[Start]" token as the initial input $P(y_1 = Astronomy|y_0 = [Start])$?

-1

Start → 0^{th} row

$k = 3$

No, the answer is incorrect.
Score: 0

Accepted Answers:

✓ (Type: Range) 0.63,0.65

2 points

astro = 0.43
char = 0.14
is = 0.09

$$\begin{array}{r} 0.43 \\ +0.14 \\ +0.09 \\ \hline 0.66 \end{array}$$

$$ast = \frac{0.43}{0.66} = 0.65$$

0 → astro
1 → is
2 → a
3 → M ⇒ chara(2)
4 → build
5 → M ⇒ exp (7)

5) Let us consider the vocabulary $\mathcal{V}$ given below.

$\mathcal{V}$ =([CLS], building, character, a, is, astronomy, science, experience, natural,[SEP]).

Now we train the BERT model by randomly initializing the parameters. Suppose we feed in the following input sequence ``Astronomy is a [Mask] building [Mask] ", where we replaced the words ``character" and ``experience" with the special [Mask] token. We just feed this single sentence (segment) to the model and do Masked Language Modelling (MLP). Note that the special tokens [CLS] and [SEP] were not used. The matrix below gives the prediction probabilities for the given input sequence

$$\begin{array}{c c} 0 & \\ 1 & \\ 2 & \\ 3 & \\ 4 & \\ 5 & \end{array} \hat{Y} = \begin{bmatrix} 0.05 & 0.02 & 0.41 & 0.07 & 0.09 & 0.14 & 0.08 & 0.05 & 0.08 & 0.01 \\ 0.10 & 0.04 & 0.06 & 0.16 & 0.04 & 0.01 & 0.06 & 0.09 & 0.01 & 0.43 \\ 0.05 & 0.07 & 0.28 & 0.03 & 0.29 & 0.08 & 0.04 & 0.08 & 0.04 & 0.04 \\ 0.10 & 0.02 & 0.14 & 0.02 & 0.01 & 0.07 & 0.06 & 0.48 & 0.07 & 0.01 \\ 0.08 & 0.29 & 0.04 & 0.01 & 0.04 & 0.03 & 0.25 & 0.05 & 0.15 & 0.06 \\ 0.14 & 0.13 & 0.22 & 0.08 & 0.01 & 0.03 & 0.06 & 0.23 & 0.08 & 0.03 \end{bmatrix}$$

char ⇒ $p = 0.14$

$L = -\ln(0.14) = 1.9661$

exp → $p = 0.23$

$L = -\ln(0.23) = 1.4697$

What is the total loss value incurred by the model for the given input sequence? Use natural logarithm where required.

3.4

Yes, the answer is correct.
Score: 2

Total loss = 3.4358

Accepted Answers:

(Type: Range) 3.3,3.5

2 points

6) Suppose we use the pre-trained BERT model for a sentiment classification problem. Assume that we use the model for feature extraction. That is, the model gives the representation for the given input sentence and we use that as input to some traditional classification algorithms. Then choose which of the following statements is (are) True **2 points**

- ✓ We can only take the representation of [CLS] token as an input to the classifier
- ○ We can also use the representation of [SEP] token as an input to the classifier
- ○ We can take representation of any word in the input sequence as an input to the classifier

in std. bert we use cls to start the seg. is used & is fed into the input as well.